Null Values and Regression

Lab 2

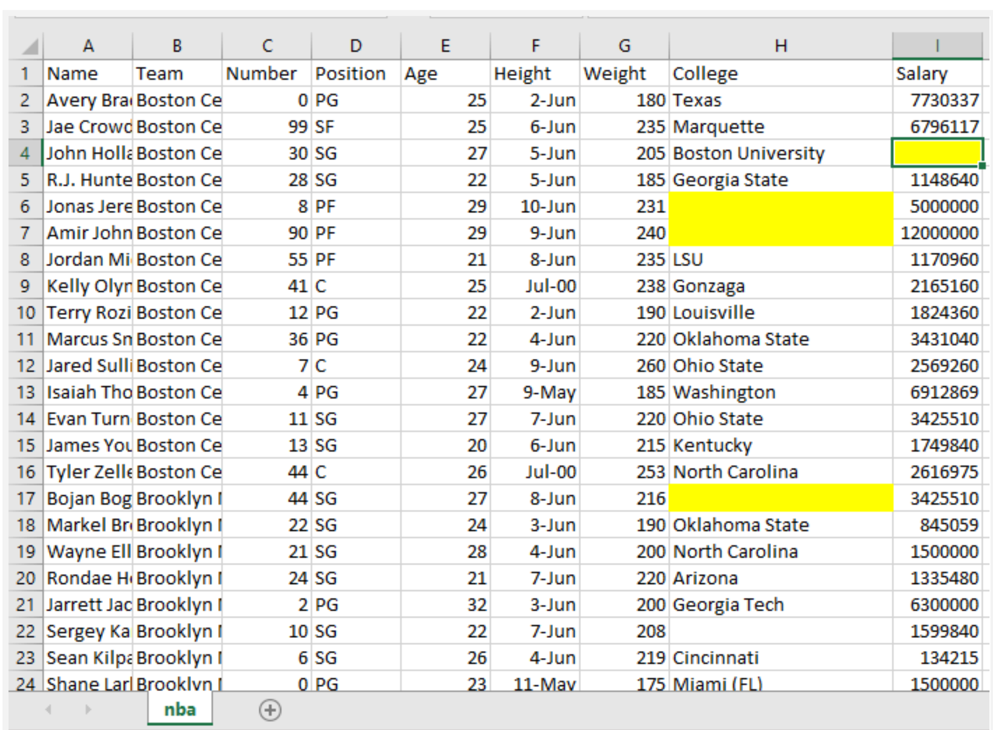
Instructions

This lab assignment exposes you to working with null values, which is data that is missing or is not known within a data set. We will analyze two different data sets for the lab to learn how to deal with null values and regression. There are four main steps:

1. Access Google Colab by signing into Google
2. Create a new notebook and install and import Python Packages
3. Download dataset Part A and summarize data
4. Download dataset Part B and summarize data

Lab Background Part A

For part A of the lab, we will be using the dataset "nba.csv", National Basketball Association, that is a data set with 458 rows and 9 column variables. The variable meta data dictionary is clear by the column heading such as name, team, number, position, etc. The data set has missing values, highlighted in yellow, as shown as a sample figure below.



	A	B	C	D	E	F	G	H	I
1	Name	Team	Number	Position	Age	Height	Weight	College	Salary
2	Avery Bra	Boston Ce	0	PG	25	2-Jun	180	Texas	7730337
3	Jae Crowd	Boston Ce	99	SF	25	6-Jun	235	Marquette	6796117
4	John Holla	Boston Ce	30	SG	27	5-Jun	205	Boston University	
5	R.J. Hunte	Boston Ce	28	SG	22	5-Jun	185	Georgia State	1148640
6	Jonas Jere	Boston Ce	8	PF	29	10-Jun	231		5000000
7	Amir John	Boston Ce	90	PF	29	9-Jun	240		12000000
8	Jordan Mi	Boston Ce	55	PF	21	8-Jun	235	LSU	1170960
9	Kelly Olyn	Boston Ce	41	C	25	Jul-00	238	Gonzaga	2165160
10	Terry Rozi	Boston Ce	12	PG	22	2-Jun	190	Louisville	1824360
11	Marcus Sn	Boston Ce	36	PG	22	4-Jun	220	Oklahoma State	3431040
12	Jared Sulli	Boston Ce	7	C	24	9-Jun	260	Ohio State	2569260
13	Isaiah Tho	Boston Ce	4	PG	27	9-May	185	Washington	6912869
14	Evan Turn	Boston Ce	11	SG	27	7-Jun	220	Ohio State	3425510
15	James You	Boston Ce	13	SG	20	6-Jun	215	Kentucky	1749840
16	Tyler Zelle	Boston Ce	44	C	26	Jul-00	253	North Carolina	2616975
17	Bojan Bog	Brooklyn I	44	SG	27	8-Jun	216		3425510
18	Markel Br	Brooklyn I	22	SG	24	3-Jun	190	Oklahoma State	845059
19	Wayne Ell	Brooklyn I	21	SG	28	4-Jun	200	North Carolina	1500000
20	Rondae H	Brooklyn I	24	SG	21	7-Jun	220	Arizona	1335480
21	Jarrett Jac	Brooklyn I	2	PG	32	3-Jun	200	Georgia Tech	6300000
22	Sergey Ka	Brooklyn I	10	SG	22	7-Jun	208		1599840
23	Sean Kilpa	Brooklyn I	6	SG	26	4-Jun	219	Cincinnati	134215
24	Shane Larl	Brooklyn I	0	PG	23	11-May	175	Miami (FL)	1500000

nba

Figure 1: Sample view of "nba.csv" data set

You will need to do something about the null values. There are several popular choices when dealing with null values:

- Eliminate the rows: A great approach if null values are a very small percentage, such as 1% of the total dataset.
- Replace with a significant value, such as the median or the mean: A great approach if the rows are valuable, and the column is reasonably balanced.

Step One:

Access Google Colab at: https://colab.research.google.com/. Select the blue 'Sign in' button from the top right corner of the page to login.

Note: *You will need a google account to be able to utilize the cloud platform.*

Step Two:

In Google Colab, select *File* from the top navigation bar and then *New notebook* to be able to install the Python Packages required for the lab.

- Rename your notebook as YourLastName_FirstName_LAB#_part# (example: Daniels_Jerome_LAB1_partA).

- Save a copy in your Drive by selecting 'File' and then 'Save a copy in Drive.'

Python Packages

```
import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
#add more python packages, as you need to do the work
```

Step Three:

Use Python in Google Colab to download your dataset "nba.csv", to explore and summarize the data as follows:

1. Find values and columns in the dataset with null values.
2. Use Pandas include a method, **fillna**, which can be used to replace null values.
3. After eliminating all null values, the dataset is much cleaner. Check that all null values have been replaced: by running code: **df.info()**

Lab Background Part B

For part B lab, we will be using the dataset "bikes.csv", which contains the hourly and daily count of rental bikes between 2011 and 2012 in the Capital bikeshare system with the corresponding weather and seasonal information. This data set has 10886 data rows for renting bikes with the data counts based on related factors such as weather, humidity, temperature, and wind speed. The data sets include 12 numerical variables and one variable as a date. As seen in the metadata dictionary below:

datetime: date and time

season: (1: springer, 2: summer, 3: fall, 4: winter)

holiday : (0: holiday, 1 is not holiday)

workingday: (if day is neither weekend nor holiday is 1, otherwise is 0.)

weather:

- 1: Clear, Few clouds, Partly cloudy, Partly cloudy
- 2: Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist
- 3: Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds
- 4: Heavy Rain + Ice Pallets + Thunderstorm + Mist, Snow + Fog

temp: Normalized temperature in Celsius.

The values are derived via (t-t_min)/(t_max-t_min), t_min=-8, t_max=+39 (only in hourly scale)

atemp: Normalized feeling temperature in Celsius.

The values are derived via (t-t_min)/(t_max-t_min), t_min=-16, t_max=+50 (only in hourly scale)

humidity: Normalized humidity. The values are divided to 100 (max)

windspeed: Normalized wind speed. The values are divided to 67 (max)

casual: count of casual users

registered: count of registered users

count: count of total rental bikes including both casual and registered

Figure 2: Metadata dictionary for "bikes.csv"

Step One:

Access Google Colab at: https://colab.research.google.com/. Select the blue 'Sign in' button from the top right corner of the page to login.

Note: *You will need a google account to be able to utilize the cloud platform.*

Step Two:

In Google Colab, select *File* from the top navigation bar and then *New notebook* to be able to install the Python Packages required for the lab.

- Rename your notebook as YourLastName_FirstName_LAB#_part# (example: Daniels_Jerome_LAB1_partB).
- Save a copy in your Drive by selecting 'File' and then 'Save a copy in Drive.'

Python Packages

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import scipy.stats as stats
import seaborn as sns
from matplotlib import rcParams
from statsmodels.formula.api import ols
from sklearn.linear_model import LinearRegression
#add more python packages, as you need to do the work
```

Step Three:

Use Python in Google Colab to download your dataset "bikes.csv", to explore and summarize the data as follows:

- **Regression using OLS Model -** Predicted for count:'total_rentals'
  - count=f('temp')
  - count=f('temp','windspeed', 'registered')
- **Regression using sklearn Model -**Predicted for count:'total_rentals'
  - count=f(temp)

Rubric

Refer to your Canvas lab assignment for the rubric.

Submission

After you complete each assignment, you will need to share your completed work by uploading an 'ipynp' and 'PDF' file of your work for your assignment submission. Follow the directions as outlined in the [Colab Setup Instructions] and follow the sharing section.

1. Name your Colab file: “YourLastName_FirstName_LAB2.ipynp”
2. Show all work as a PDF: “YourLastName_FirstName_LAB2.pdf”

Note: *You can use the same files for both of the data sets.*

Reference

Fanaee-T, Hadi. UCI. (2013). Machine Learning Repository. Bike Sharing Dataset Data Set.